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S2, S3, and S4, or jumpers in their place to save cost. Transistor T1 works as emitter-follower because at low currents, the Zener diode has high internal resistance.

The circuit produces two outputs—Output 1 for the white noise buffered from T1, and Output 2 for the amplified white noise from transistors T2 and T3. Pot VR1 is used to adjust amplitude of the noise that is amplified by T2 and T3.

Transistor T2 works as voltage amplifier, increasing the amplitude of the noise. The voltage gain depends mainly on the resistors connected to the emitter and collector of T2. The voltage at collector of T2 (point A) should be around half of the power supply voltage. To obtain that you may need to change the value of feedback resistor R10. Transistor T3 works as emitter-follower, providing low output resistance at Output 2.

The maximal amplitude depends on the Zener diode and the power supply. The power supply (V_{cc}) should be within the range of 12V to 25V. Higher V_{cc} is better as it can produce higher output signal in case of need. The circuit will work well even with 9V power supply but the Zener diode breakdown voltage rating in that case should be below 6V.

(During testing at EFY Lab, a 15V power supply and a 6.8V Zener diode were used.)

Construction and testing

An actual-size PCB layout for this simple white noise generator is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on PCB. Connect 15V DC supply to the PCB at connector CON1. Connect switches S1 through S4 to the PCB using suitable wires. The two outputs are available at CON2 and CON3 connector points. **EFY**

PARTS LIST

Semiconductors:

- T1-T3 - BC547 npn transistor
- LED1 - 5mm LED
- ZD1 - 6.8V Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R10 - 1-mega-ohm
- R2 - 470-kilo-ohm
- R3, R13, R15 - 200-kilo-ohm
- R4 - 100-kilo-ohm
- R5, R7, R14 - 300-ohm
- R6 - 100-ohm
- R8 - 6.8-kilo-ohm
- R9 - 20-ohm
- R11 - 200-ohm
- R12 - 1-kilo-ohm
- R16, R17 - 2-kilo-ohm
- VR1 - 10-kilo-ohm pot

Capacitors:

- C1, C8 - 1nF ceramic disk
- C2, C4, C6, C9, C12 - 100nF ceramic disk
- C3 - 470 μ F, 25V electrolytic
- C5, C7, C11 - 100 μ F, 25V electrolytic
- C10 - 1000 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- S1-S4 - On/off switch
- 2-pin connector for on/off switches



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